

Best-fit lines, outliers and conclusions

Working scientifically
Both papers · Combined and Separate
10 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	What is a line of best fit?	One smooth line showing the overall trend of the data.	☐
2 ★	Describe how a line of best fit should be drawn.	As one thin line with points balanced above and below it.	☐
3	State what must not be done when drawing a line of best fit.	Joining the points dot-to-dot.	☐
4 ★	What is an outlier?	A result that does not fit the pattern of the others.	☐
5 ★	Describe what you should do when you find an outlier.	Check the recording and plotting, then repeat the measurement. Never just delete it.	☐
6	What is a trend?	The overall relationship shown by the data.	☐
7	State what a good conclusion must include.	The relationship between the variables, supported by values with units.	☐
8 ★	Describe the graph of two quantities that are directly proportional.	A straight line through the origin.	☐
9	State the three parts of a good evaluation point.	The problem, its effect on the data, and an improvement that fixes it.	☐
10	What is peer review?	When other scientists check the method, the evidence and the claims.	☐